

CANDIDATE NAME _____

INDEX NUMBER _____

CG _____



SERANGOON JUNIOR COLLEGE
JC2 Preliminary Examination 2014

H2 BIOLOGY
9648

Paper 2

2 Hours

Additional materials:
Writing papers / OMS

Date / Day: Tuesday 19th August 2014**INSTRUCTIONS TO CANDIDATES**

Write your name, CG and index number in the spaces at the top of this page and on all separate writing papers used.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Section A

Answer **all** questions.

Write your answers in spaces provided on the question paper.

Section B

Answer **only one** question out of two.

Write your answers on the separate answer paper provided.

INFORMATION FOR CANDIDATES

The intended number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE	
Section A	
1	/11
2	/10
3	/11
4	/11
5	/7
6	/10
7	/14
8	/6
Section B	
9	/20
10	/20
TOTAL	/100

This question paper consists of 20 printed pages

Question 1

Porins are transmembrane proteins found on the outer membrane of mitochondria. Their function is to transport substances across the membrane. **Fig. 1.1a** and **Fig. 1.1b** show the side view and top view of the structure of a single polypeptide of porin respectively.

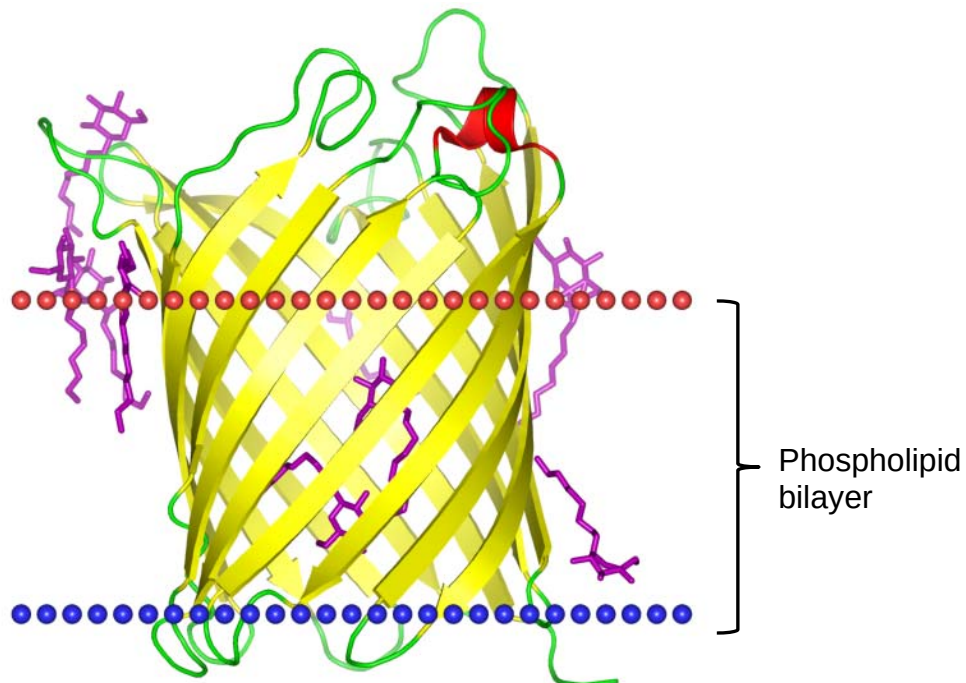


Fig. 1.1a



Fig. 1.1b

(a) Name a covalent bond present within the protein porin. [1]

.....

(b) With reference to **Fig. 1.1a** and **Fig. 1.1b**,

(i) Name a non-covalent bond present within the protein porin. [1]

.....

(ii) Describe the structure of the protein porin. [2]

.....

.....

.....

.....

A researcher intends to study the transport of substances carried out by porins across the outer mitochondrial membrane. The researcher begins by isolating mitochondria from rat liver cells and placing them in 5 different solutions, each containing a different substance with a concentration of 10 arbitrary units as shown in **Table 1.2**.

Table 1.2

Solution	Content of Solution
1	10 AUs of Na ⁺ ions
2	10 AUs of Cl ⁻ ions
3	10 AUs of H ⁺ ions
4	10 AUs of protein insulin
5	10 AUs of acidic amino acid glutamate

The concentrations of the substances in the intermembrane space of the mitochondria were measured before the experiment and after 15 minutes of being placed in the solutions. The results obtained by the researcher are shown in **Table 1.3** on the following page (page 4).

Table 1.3

Solution	Content of Solution	Concentration of substances in mitochondrial intermembrane space before experiment	Concentration of substances in mitochondrial intermembrane space after 15 minutes
1	10 AUs of Na ⁺ ions	1 AU of Na ⁺ ions	10 AUs of Na ⁺ ions
2	10 AUs of Cl ⁻ ions	2 AUs of Cl ⁻ ions	10 AUs of Cl ⁻ ions
3	10 AUs of H ⁺ ions	2 AUs of H ⁺ ions	10 AUs of H ⁺ ions
4	10 AUs of protein insulin	0 AU of protein insulin	0 AU of protein insulin
5	10 AUs of acidic amino acid glutamate	0 AU of acidic amino acid glutamate	9 AUs of acidic amino acid glutamate

(c) With reference to **Table 1.2** and **Table 1.3**,

- (i) Name the process by which the substances are transported through porins. [1]

.....

- (ii) Explain why insulin is unable to enter the mitochondrial intermembrane space. [2]

.....

.....

.....

(d) Porins were artificially inserted into the membranes of a motor neurone. Predict and explain the effect of this modification on the function of the neurone. [4]

.....

.....

.....

.....

.....

.....

[Total: 11 marks]

Question 2

Scientists often employ the method of fluorescence imaging to study the process of nuclear division in cells. This method involves the use of fluorescence tagged antibodies to label specific structures within the cell, so that their movements and behaviours can be followed over the course of nuclear division.

A group of SR scientists decided to employ the aforementioned method to study the changes in centromere number in an animal cell undergoing meiosis. Initial testing showed that each centromere could be visualised as a single dot and therefore the total number of dots can be taken to correspond to the total number of centromeres within the cell. **Table 2.1** shows the number of dots and hence centromeres observed per cell, at various stages of meiosis.

Table 2.1

Stage of Meiosis	Prophase I	Prophase II	Anaphase II	Telophase II
Number of dots observed per cell	16			16

(a) With reference to **Table 2.1**,

- (i) Predict the number of dots (i.e. centromeres) observed in prophase II and anaphase II respectively. [2]

Prophase II:

.....

Anaphase II:

.....

- (ii) Explain your answer to part **(a) (i)**. [2]

Prophase II:

.....

.....

Anaphase II:

.....

.....

When the same experiment was conducted while the cell was in interphase, prior to meiosis I, no distinct dots could be observed in the nucleus. Instead, the entire nucleus emitted fluorescence.

(b) Suggest a reason for this observation. [1]

.....

.....

Apart from using fluorescence imaging to determine the changes in centromere numbers, this technique was also employed by the SR scientists to study the movement of chromosomes during meiosis in the animal cell. More specifically, the changes in distance between 2 corresponding regions of non-sister chromatids of a pair of homologous chromosomes (**X**) were measured over the course of the first meiotic division. The results are shown in **Fig 2.1** below.

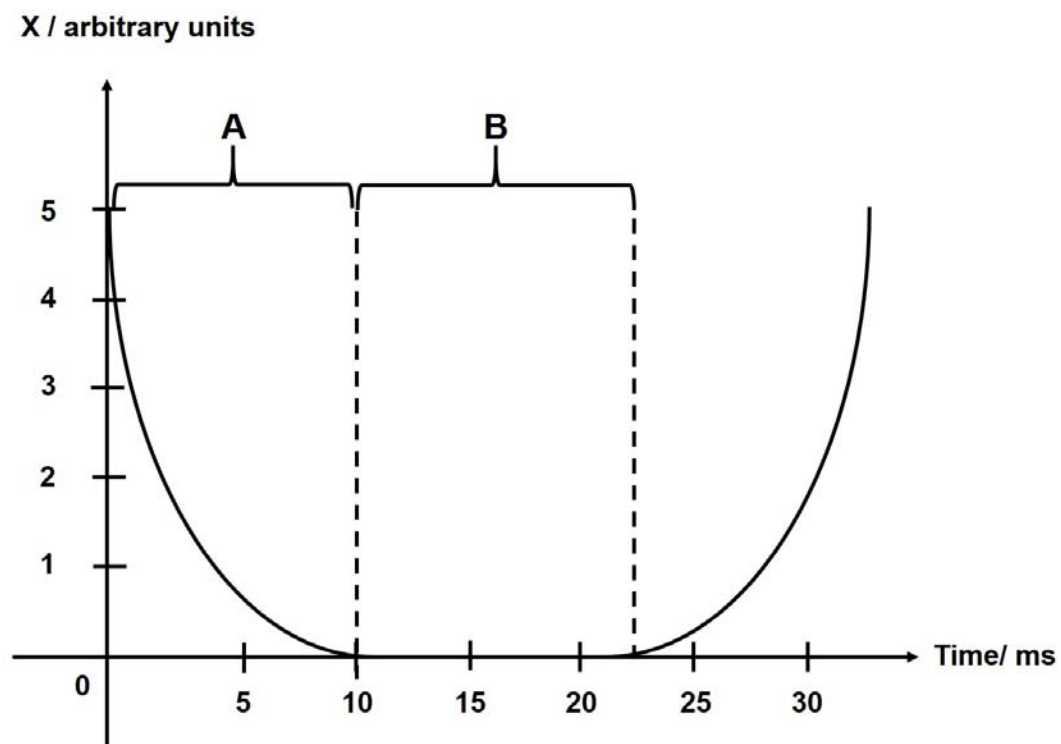


Fig 2.1

(c) With reference to **Fig 2.1**, explain the changes in **X** for

(i) Region A. [2]

.....

.....

.....

.....

(ii) Region B. [2]

.....

.....

.....

.....

Driven by the success of the above experiment, one of the SR scientists decided to conduct the same experiment for the second meiotic division of the animal cell. His investigations however did not produce any results.

(d) State a reason for the lack of results obtained. [1]

.....

.....

[Total: 10 marks]

Question 3

Fig. 3.1 depicts a cell's responses to two kinds of DNA damage, a single stranded break and a double stranded break.

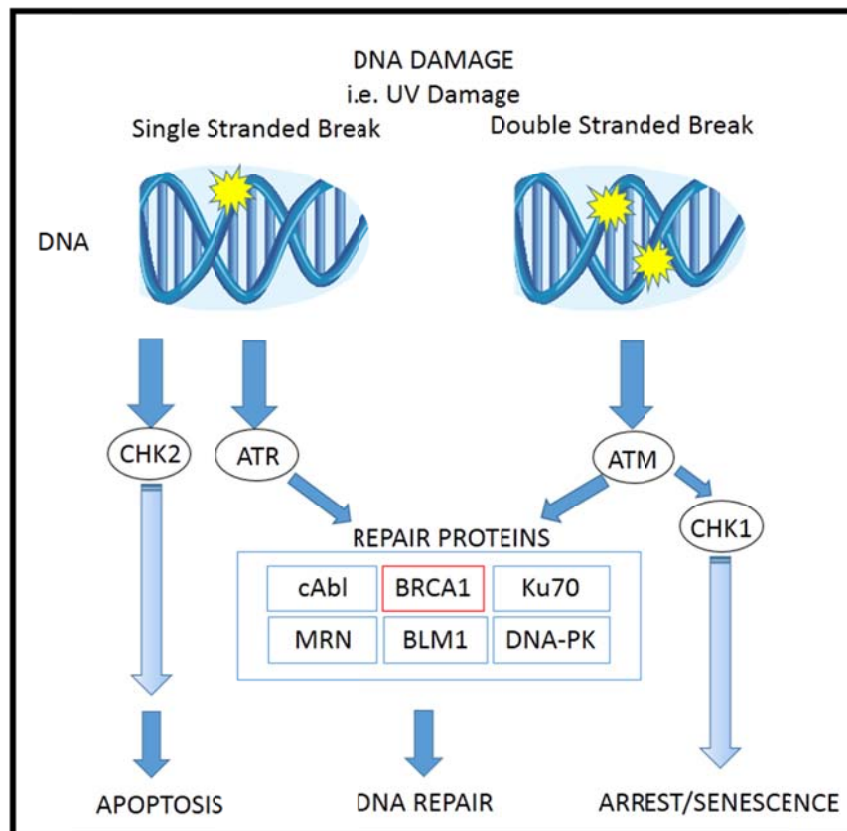


Fig. 3.1

(a) State the bond that would need to be broken to cause both types of DNA damage. [1]

.....

(b) With reference to **Fig. 3.1**, describe the sequence of events that occurs in a cell that has a single strand break, eventually leading to DNA repair. [3]

.....

.....

.....

.....

.....

.....

(c) (i) Name the class of genes that produce the repair proteins. [1]

.....

(ii) Discuss the significance of this class of genes to cells. [2]

.....

.....

.....

.....

(d) Describe one class of substances that can cause DNA damage and explain how they bring about the damage. [2]

.....

.....

.....

.....

.....

.....

(e) Suggest a method by which cells will be able to repair a single strand break and explain why a double strand break may be harder to repair. [2]

.....

.....

.....

.....

.....

.....

[Total: 11 marks]

Question 4

Fig. 4.1 shows a plasmid commonly used to transfer genes to another bacterial cell via conjugation. This is a method for cloning genes into bacterial cells by using the plasmid of another bacterial cell as a vector. For conjugation to occur, the bacterial strain that passes the plasmid must have the genes necessary to build the conjugative apparatus (a pilus) between the bacterial cells.

This is normally carried on a large plasmid, called the F plasmid, that itself is transferred during conjugation, but due to its large size, it is not amenable to many such cloning strategies. In an attempt to still carry out cloning via conjugation in the laboratory without using the F plasmid, a small conjugative plasmid containing the origin sequence for conjugative transfer was created. However, this plasmid, as shown in **Fig. 4.1**, still does not contain the necessary genes to build the pilus.

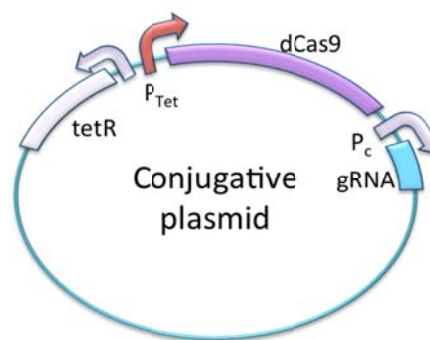


Fig. 4.1

(a) Describe the process of conjugation and elaborate on its significance. [4]

.....

.....

.....

.....

.....

.....

.....

(b) Describe how the F plasmid differs from the conjugative plasmid in **Fig. 4.1**. [3]

.....

.....

.....

.....

.....

.....

(c) As noted above, the conjugative plasmid does not have the required genes to produce the pilus. Assuming you only have this plasmid to use, suggest how you could overcome this issue and successfully transfer a gene on the conjugative plasmid via conjugation to another bacterial cell. [2]

.....

.....

.....

.....

.....

.....

(d) Suggest a use for the tetR gene on the conjugative plasmid. [2]

.....

.....

.....

.....

[Total: 11 marks]

Question 5

Fig. 5.1 below shows the ears of maize (*Zea mays*). On the left, is variety Tom Thumb, the Black Mexican one on the right, and that of their F₁ hybrid in the centre.

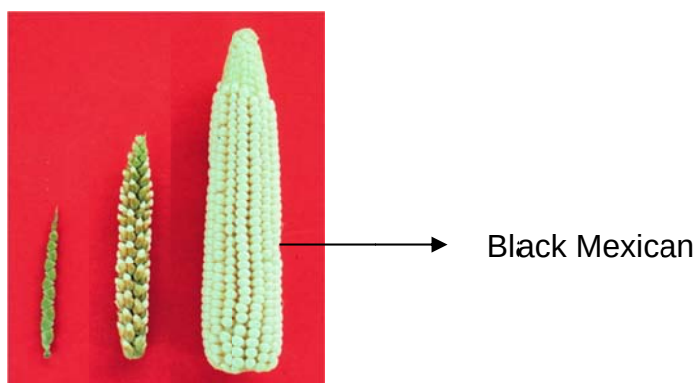


Fig. 5.1

(photo by John Doebley)

A cross was made between 2 pure breeding maize varieties, Tom Thumb and Black Mexican which differed markedly in ear length. The ear length of both parents, the F₁ and the F₂ generations, was measured in centimetre. This is given in the table below with the number of ears in each length category (e.g.14 of the F₁ plants produced ears 12 cm in length).

	Ear length (cm)																
	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Black Mexican parents									3	11	12	15	26	15	10	7	2
Tom Thumb parents	4	21	24	8													
F₁					1	12	12	14	17	9	4						
F₂			1	10	19	26	47	73	68	68	39	25	15	9	1		

Both parents showed variations in ear length. The variations in the F₁ are comparable to the average of the parental variations while the variations in the F₂ are greater than that found within the parental lines or the F₁.

(a) Suggest an explanation for this. [3]

.....

.....

.....

.....

.....

.....

(b) Explain the genetic basis behind the wide range of ear lengths. [3]

.....

.....

.....

.....

.....

.....

(c) What is the term used to describe the range of phenotypes in the above example? [1]

.....

[Total: 7 marks]

Question 6

In a college lab deep in opposition territory, an experiment was carried out using yeast cells (*Saccharomyces cerevisiae*) in anaerobic conditions. **Yeasts** are eukaryotic microorganisms classified in the kingdom Fungi, with 1,500 species currently described (estimated to be 1% of all fungal species). Yeasts are unicellular organisms, although some species with yeast forms may become multicellular. The experiment was set up as illustrated in the **Fig. 6.1**.

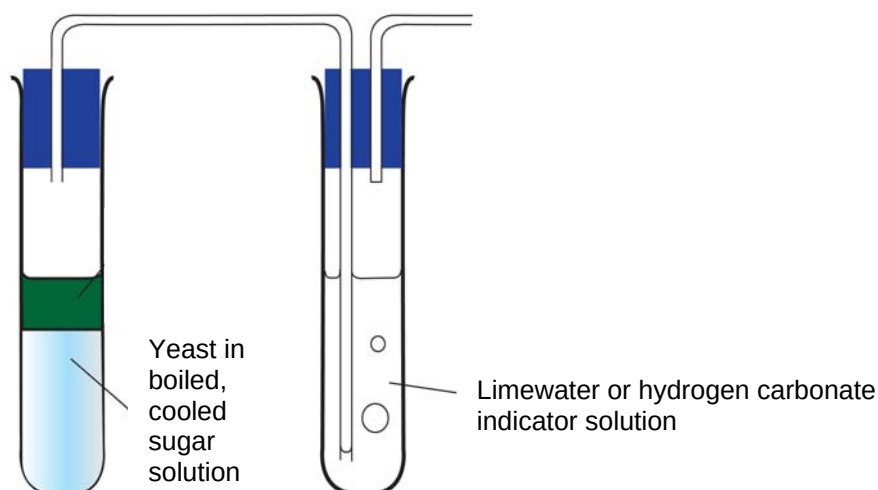


Fig. 6.1

- (a) During the experiment, it was observed that the lime water turned cloudy. Explain how the yeast cells were able to bring about the cloudiness in the lime water. [3]

.....

.....

.....

.....

.....

.....

- (b) A very conscientious student decided to do a follow-up experiment. He used the exact same experimental setup but replaced the yeast cells with cells from a mouse cell line. He found that the lime water did not turn cloudy. Comment on this observation. [2]

.....

.....

.....

.....

In a separate experiment, the respiration rate of yeast was measured using a variety of substrates. The results are shown in **Fig. 6.2** below.

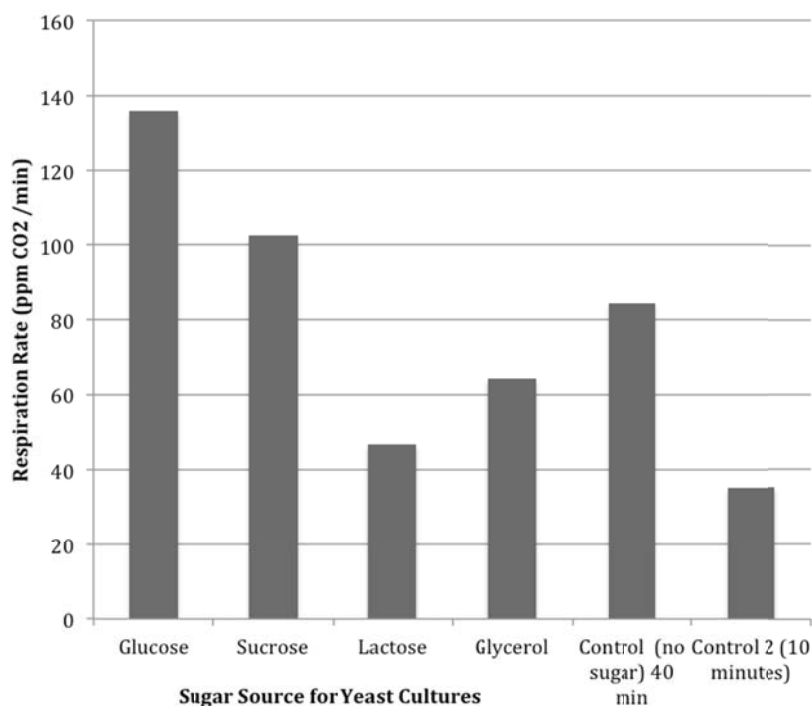


Fig. 6.2

(c) Explain why glucose appears to be a more effective respiratory substrate than sucrose. [3]

.....

.....

.....

.....

.....

.....

(d) Suggest why using lactose as a substrate actually results in a lower respiration rate than control 1, which does not have any sugar provided. [2]

.....

.....

.....

.....

[Total: 10 marks]

Question 7

Fig. 7.1 shows the changes in beak size of medium ground finches on the Galapagos Islands over a period of about 8 years.

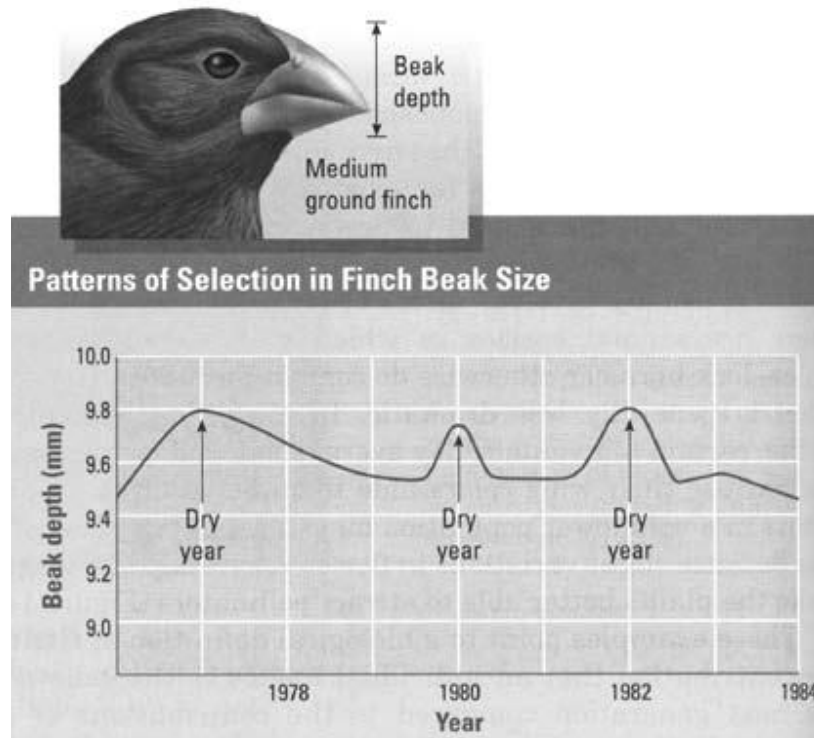


Fig. 7.1

- (a) (i) Describe the changes in beak size exhibited by the finches as reported in **Fig. 7.1**. [2]

.....

.....

.....

.....

- (ii) Suggest reasons for the changes described. [2]

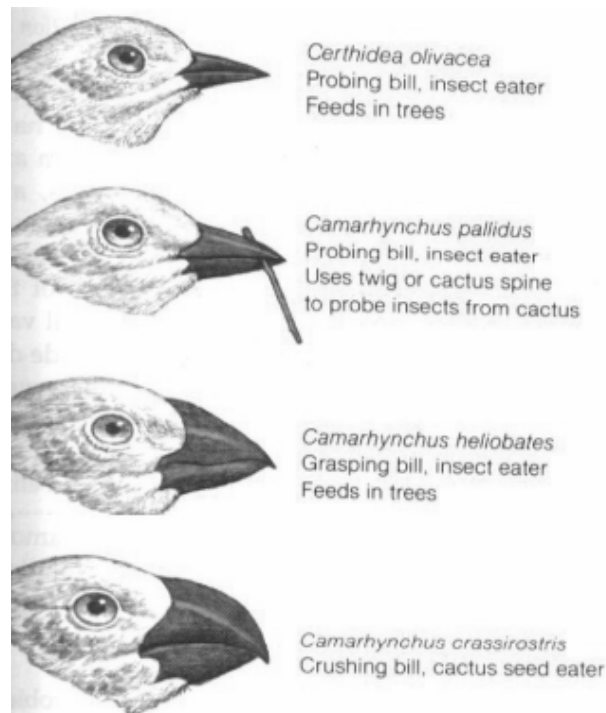
.....

.....

.....

.....

When scientists explored one of the islands, the following species of finches were observed.



(b) State the number of species of finches found on this island. [1]

.....

It is believed that *Camarhynchus pallidus* is the original species to inhabit the island, having been forcibly blown in from the mainland during a particularly violent storm.

(c) Explain how this original ancestral species could have evolved into the current group of finch species found on the island. [4]

.....

.....

.....

.....

.....

.....

.....

Fig. 7.2 below shows the relationships between a number of finches including *Camarhynchus pallidus*. This relationship diagram was constructed using molecular data.

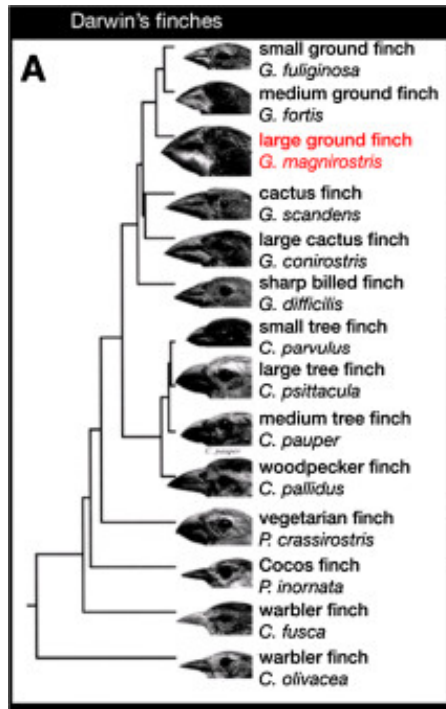


Fig. 7.2

(d) Name this relationship diagram and briefly explain how it could have been constructed. [3]

.....

.....

.....

.....

.....

.....

(e) Describe the advantages of using molecular data in constructing this kind of diagram. [2]

.....

.....

.....

.....

[Total: 14 marks]

Question 8

Multiple sclerosis (MS) involves an immune-mediated process in which an abnormal response of the body's immune system is directed against the central nervous system (CNS), which is made up of the brain, spinal cord and optic nerves. Within the CNS, the immune system attacks myelin — the fatty substance that surrounds and insulates the nerve fibers — as well as the nerve fibers themselves. **When any part of the myelin sheath or nerve fiber is damaged or destroyed, nerve impulses traveling to and from the brain and spinal cord are distorted or interrupted**, producing a wide variety of symptoms.

Adapted from The National Multiple Sclerosis Society Website

(a) Describe briefly the structure of a myelinated neurone. [2]

.....

.....

.....

.....

(b) Describe the normal function of a myelin sheath in nerve cells. [2]

.....

.....

.....

.....

(c) Suggest how damage to the myelin sheath can lead to nerve impulses traveling to and from the brain and spinal cord becoming distorted or interrupted. [2]

.....

.....

.....

.....

[Total: 6 marks]

Section B (20 Marks)

Answer **one** question

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.

Your answers must be in continuous prose where appropriate.

Your answers must be set out in sections (a), (b), etc as indicated in the question.

Question 9

- (a)** Explain the principles of homeostasis in terms of receptors, effectors and negative feedback. [7]
- (b)** Compare competitive and non-competitive inhibitors and their effects on the rate of enzyme activity. [7]
- (c)** Suggest similarities between the interaction of substrate with an enzyme and the interaction of ligand with a receptor. [6]

Question 10

- (a)** Using named examples, compare repressible and inducible systems of gene regulation. [10]
- (b)** Explain how the prokaryotic genome structure differs from eukaryotes. [6]
- (c)** Discuss the evolutionary significance of organizing genes into chromosomes. [4]